Chem 250L – Spectrophotometers and Optical Density: A Protein-Powder Turbidity Standard Curve

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September 29, 2026

Abstract

A protein-powder suspension was prepared, diluted serially by factors of two, and measured at 595 nm with a SpectroVis Plus spectrophotometer to build a turbidity standard curve. Five independent trials each gave five standards from $0.005\,\mathrm{g\,mL^{-1}}$ to $0.000\,312\,5\,\mathrm{g\,mL^{-1}}$. The mean optical densities were 0.9654, 0.5664, 0.3348, 0.1838, and 0.0962, with relative standard deviations of 3–9%. A least-squares fit of the mean values gave $\mathrm{OD}_{595} = 182.19\,C + 0.0763$ ($R^2 = 0.9917$; C in $\mathrm{g\,mL^{-1}}$), and the five individual-trial slopes agreed to within 2.7% RSD. Although R^2 is high, a residual analysis shows a systematic curvature: OD rises as $C^{0.83}$ rather than in direct proportion to C, the intercept is significantly nonzero, and the linear equation recovers the lowest standard with only 35% accuracy. The curve is therefore reliable for estimating concentrations between about 0.0006 and $0.005\,\mathrm{g\,mL^{-1}}$ but not below that range. No unknown sample was measured, so no unknown concentration is reported.

1 Introduction

Spectrophotometry measures how a sample interacts with light of a chosen wavelength. A spectrophotometer sends light through a blank or sample, measures the transmitted intensity, and compares it with the blank. In this experiment a SpectroVis Plus instrument was used at 595 nm to measure the optical density (OD) of protein-powder suspensions that served as a safe model for a growing bacterial culture. As particle concentration increases, the suspension becomes cloudier and fewer photons reach the detector, so the OD reading increases; this is how culture growth is commonly monitored.

Unlike a clear solution of one absorbing species, a suspension attenuates light mainly by *scattering*. OD therefore measures turbidity and relative particle concentration, not a molar quantity, and the OD–concentration relationship is an empirical calibration rather than a direct application of the Beer–Lambert law. The goals here were to prepare accurate standards by serial dilution, measure their OD, and evaluate how well a linear standard curve describes the response.

2 Background and Theory

2.1 Optical Density and Light Scattering

The instrument reports a quantity based on the transmitted intensity I relative to the blank intensity I_0:

$$\mathrm{OD} = -\log_{10}\left(\frac{I}{I_0}\right). \tag{1}$$

For a clear solution of a single absorber the Beer–Lambert law, $A = \varepsilon bc$, predicts that A is proportional to concentration with zero intercept. In a suspension, scattering also removes light, and at higher particle densities multiple scattering and particle shadowing mean that the response need not stay exactly proportional to concentration. A measured OD of 1.0 corresponds to only 10% of the incident light being transmitted, which is one reason the lab manual recommends keeping readings below about 1.0–1.1.

2.2 Blank, Cuvette, and Instrument Handling

The blank (distilled water) corrects for light lost to the solvent and cuvette, so the instrument reads approximately zero before any sample is measured. Cuvettes have a defined path length, commonly 1 cm; the clear faces must face the light beam and the frosted or ribbed faces are used for handling. Fingerprints, scratches, droplets, and bubbles scatter light and falsely increase the OD.

2.3 Serial Dilution

In a serial 1:2 dilution, each tube receives equal volumes of the previous solution and solvent, so every step halves the concentration:

$$C_n = C_0\left(\tfrac{1}{2}\right)^n, \qquad n = 0,1,2,3,4. \tag{2}$$

Because each tube is made from the previous one, any pipetting error propagates and accumulates down the series, so the most dilute standard carries the largest relative uncertainty.

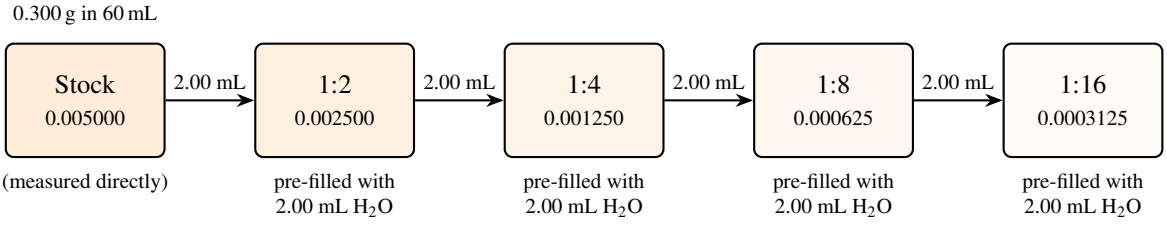


Figure 1: Serial 1:2 dilution scheme. Each tube initially contained 2.00 mL of distilled water; 2.00 mL of the previous solution was transferred and mixed. Concentrations are in $\mathrm{g\,mL^{-1}}$.

2.4 Standard Curve, Regression, and Error Statistics

A standard curve relates the measured response y (OD) to known concentrations x (C). Least squares gives

$$m = \frac{S_{xy}}{S_{xx}}, \qquad b = \bar{y} - m\bar{x}, \qquad s_{y/x} = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-2}}, \tag{3}$$

with standard errors $s_m = s_{y/x}/\sqrt{S_{xx}}$ and $s_b = s_{y/x}\sqrt{1/n + \bar{x}^2/S_{xx}}$. An unknown with a measured OD of y is quantified by

$$C_{\text{unknown}} = \frac{y - b}{m}, \tag{4}$$

multiplied by any dilution factor. The detection and quantitation limits are estimated as $\text{LOD} = 3.3\, s_{y/x}/m$ and $\text{LOQ} = 10\, s_{y/x}/m$.

3 Objectives

1. Work with a fine powder while maintaining a clean working area.
2. Weigh accurately and prepare a protein-powder suspension of known concentration.
3. Practice pipetting and 1:2 serial dilution.
4. Use a SpectroVis Plus spectrophotometer to quantify turbidity as OD_{595}.
5. Construct, analyze, and critically evaluate a standard curve from five replicate trials.

4 Hypothesis

If the protein-powder concentration increases, then OD_{595} will increase because more suspended particles will scatter and absorb light away from the detector. The relationship should be approximately linear over the tested range, so a linear regression could be used to estimate an unknown concentration.

5 Materials and Equipment

- Protein powder and distilled water
- Clean cuvettes (1 cm path) and Kimwipes
- Milligram balance, weighing vessel, and clean spatula
- Beaker or centrifuge tube for the stock suspension; labeled dilution tubes
- Micropipettes and clean tips
- SpectroVis Plus visible/near-UV spectrophotometer (operating range about 380–950 nm)

6 Safety Considerations

Safety goggles were worn at all times, and gloves and a lab coat were used as directed. The main hazard was inhalation of fine powder, so the container was kept closed except during transfer and powder was added gradually, never directly onto the balance pan. Spills were removed promptly, hands were washed as needed, and liquids were kept away from the spectrophotometer. Following the R.A.M.P. approach, the hazards were recognized, the risks assessed, exposure minimized with PPE, and emergency procedures reviewed before starting.

7 Methods and Procedure

7.1 Stock Suspension

A clean weighing vessel was tared on a milligram balance and protein powder was added with a clean spatula. The actual mass was recorded immediately (see Table 1). The powder was transferred to a clean container, distilled water was added to the recorded volume, and the mixture was stirred until the powder was dispersed and homogeneous, avoiding persistent bubbles.

7.2 Blanking

The SpectroVis Plus was turned on and allowed to initialize, absorbance/OD mode was selected, and the wavelength was set to 595 nm and verified. A clean, unscratched cuvette was held by its frosted sides, filled about two-thirds to three-quarters full with distilled water, wiped with a Kimwipe, and placed with its clear faces in the light path. After closing the lid, the instrument was zeroed and the baseline was confirmed to be approximately zero.

7.3 Serial Dilution and Measurement

Four labeled tubes (1:2, 1:4, 1:8, 1:16) were each pre-filled with 2.00 mL of distilled water. 2.00 mL of stock was added to the 1:2 tube and mixed thoroughly. Using a clean tip each time, 2.00 mL was then transferred from each tube into the next and mixed before the following transfer (Figure 1). Every sample was mixed gently immediately before measurement, inspected for bubbles, and measured with the same blank, wavelength, cuvette type, fill level, and orientation. The undiluted stock and the four dilutions were each measured, giving five standards per trial. The target OD of the stock was 0.90–1.10. The experiment was repeated for a total of five trials.

8 Results

8.1 Stock Preparation

Table 1: Recorded stock-suspension masses, volumes, and calculated concentrations. Trial numbers follow the spreadsheet; each mass is paired with its data set (see note below).

Trial	**Powder (g)**	**Water (mL)**	**Stock conc. (g mL^{-1})**	**Stock OD$_{595}$**
1	0.300	60	0.005000	0.984
2	0.300	60	0.005000	0.980
3	0.300	60	0.005000	0.980
4	0.299	60	0.004983	0.912
5	0.150	30	0.005000	0.971

All five stock suspensions had an OD within or just outside the 0.90–1.10 target range (0.912–0.984), so no additional adjustment of the stock was needed. Trial 4 used 0.299 g; the resulting 0.3% concentration difference is negligible (see Section 8.6), so the nominal concentrations of 0.005 g mL^{-1} were used for all trials, as in the spreadsheet graphs.

Note on trial numbering: the notebook records the data set beginning at 0.912 as "trial 2" and the set beginning at 0.980/0.563 as "trial 4," whereas the spreadsheet uses the opposite order. The spreadsheet numbering is used throughout this report.

8.2 Replicate OD_{595} Data

Table 2: OD_{595} of the five standards in each trial, with mean, sample standard deviation, and relative standard deviation (RSD).

	Concentration (g mL^{-1})				
Trial	0.005	0.0025	0.00125	0.000625	0.0003125
1	0.984	0.581	0.350	0.189	0.101
2	0.980	0.563	0.347	0.189	0.093
3	0.980	0.586	0.328	0.183	0.093
4	0.912	0.555	0.304	0.171	0.086
5	0.971	0.547	0.345	0.187	0.108
Mean	0.9654	0.5664	0.3348	0.1838	0.0962
SD ($n = 5$)	0.0302	0.0167	0.0192	0.0076	0.0085
RSD (%)	3.1	2.9	5.7	4.1	8.8

8.3 Individual-Trial Standard Curves

Figure 2 shows a separate scatter plot and best-fit line for each trial, with concentration on the x-axis and OD_{595} on the y-axis (a scatter plot rather than a line chart, so concentration is not treated as an evenly spaced category). Table 3 gives the regression results.

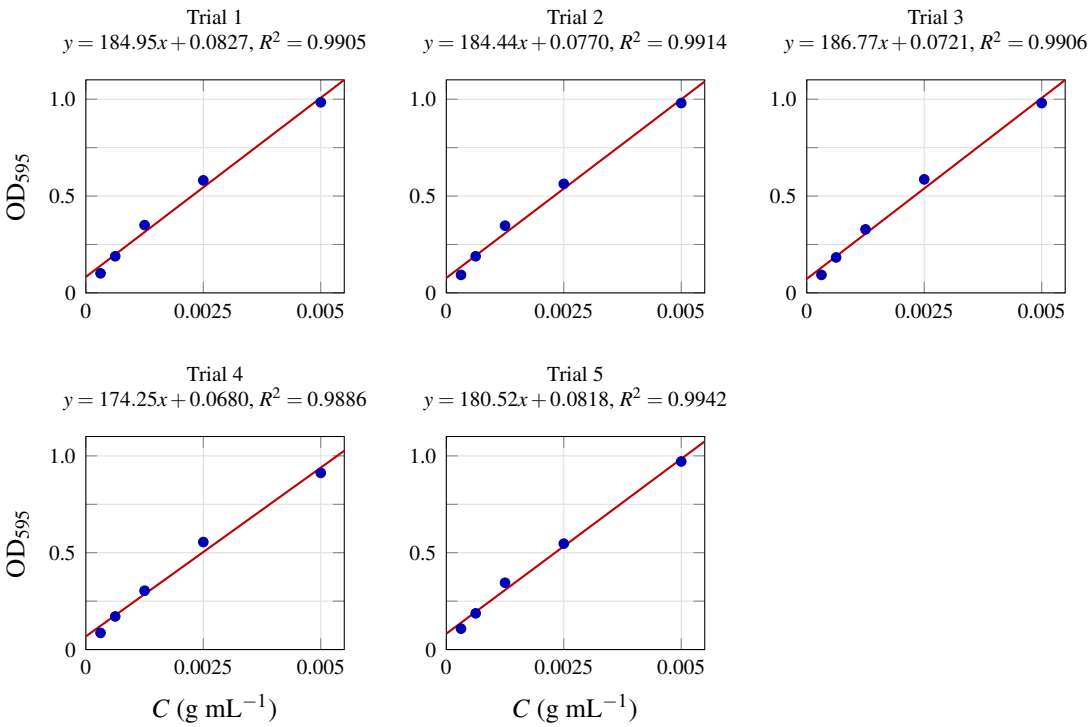


Figure 2: Standard curve for each of the five trials (blue points: measured OD_{595}; red line: least-squares fit). The lines are nearly parallel, indicating good reproducibility of the response.

Table 3: Linear regression of OD_{595} (y) on concentration (x) for each trial ($n = 5$ points each). Uncertainties are one standard error.

Trial	**Slope m (mL g^{-1})**	**Intercept b**	$s_{y/x}$	R^2
1	184.9 ± 10.5	0.083 ± 0.027	0.0400	0.9905
2	184.4 ± 9.9	0.077 ± 0.026	0.0378	0.9914
3	186.8 ± 10.5	0.072 ± 0.027	0.0401	0.9906
4	174.2 ± 10.8	0.068 ± 0.028	0.0411	0.9886
5	180.5 ± 7.9	0.082 ± 0.021	0.0303	0.9942
Mean $\pm$ SD of trials	182.2 ± 5.0	0.0765 ± 0.0058	–	0.9911

The five slopes average $182.2\,\mathrm{mL\,g^{-1}}$ with a standard deviation of 5.0 (2.7% RSD), and the intercepts agree to within about 0.015 OD units, so the calibration is reproducible from trial to trial.

8.4 Composite Standard Curve

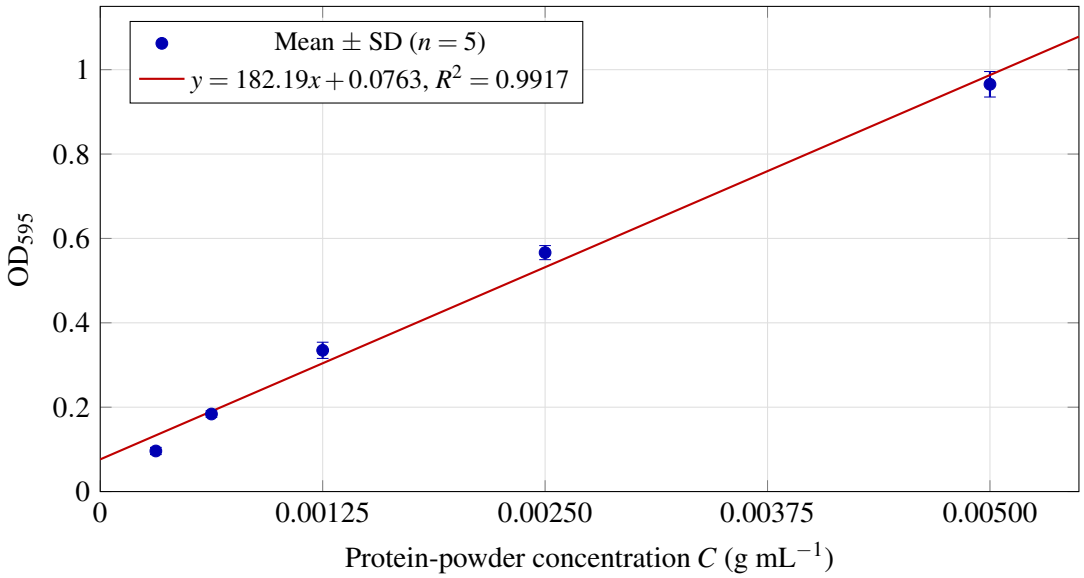


Figure 3: Standard curve from the five-trial mean OD_{595} values. Error bars are one sample standard deviation.

Regression of the mean values (Figure 3) gave

$$\boxed{\mathrm{OD}_{595} = (182.19 \pm 9.62)\,C + (0.0763 \pm 0.0248)}, \qquad R^2 = 0.9917, \quad s_{y/x} = 0.0367, \tag{5}$$

where C is in $\mathrm{g\,mL^{-1}}$ and the uncertainties are standard errors ($n = 5$ means, 3 degrees of freedom). The estimated limits are LOD $\approx 0.000\,66\,\mathrm{g\,mL^{-1}}$ and LOQ $\approx 0.0020\,\mathrm{g\,mL^{-1}}$.

8.5 Residual and Linearity Analysis

Table 4: Fit of the mean curve: residuals and concentrations back-calculated from the fitted line.

C (g mL^{-1})	Mean OD	Fitted OD	Residual	C_{calc} (g mL^{-1})	Recovery (%)
0.005000	0.9654	0.9873	−0.0219	0.004880	97.6
0.002500	0.5664	0.5318	+0.0346	0.002690	107.6
0.001250	0.3348	0.3041	+0.0307	0.001419	113.5
0.000625	0.1838	0.1902	−0.0064	0.000590	94.4
0.0003125	0.0962	0.1333	−0.0371	0.000109	34.9

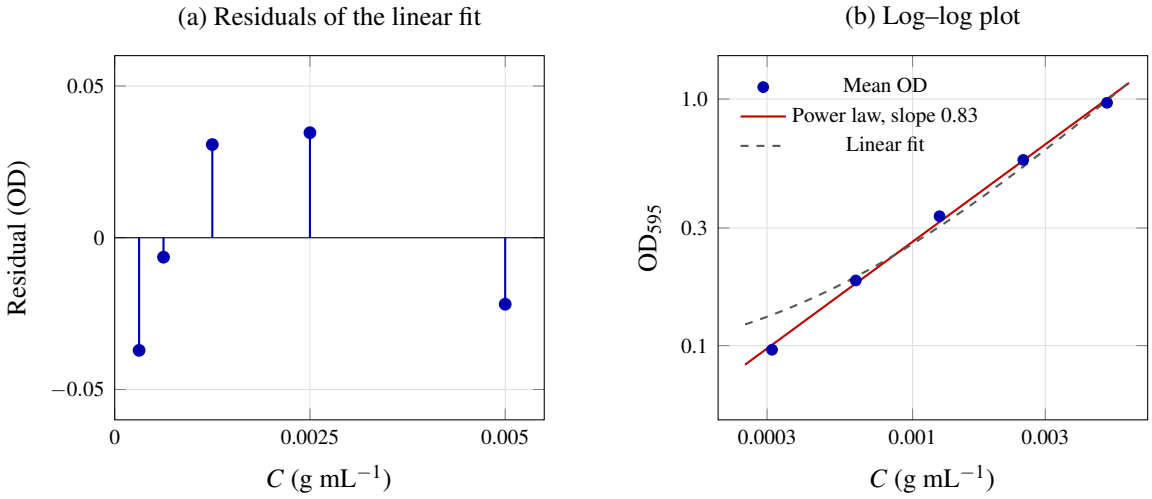


Figure 4: (a) Residuals of the linear fit to the mean data show a systematic $-,+,+,-,-$ pattern rather than random scatter. (b) On log–log axes the data follow a straight line with slope 0.83, while the linear fit (dashed) bends away from the data at low concentration.

8.6 Sensitivity of the Fit

Table 5: Effect of data-handling choices on the regression of OD on C.

Data set	**Slope (mL g^{-1})**	**Intercept**	R^2
Five trial means (reported curve)	182.19	0.0763	0.9917
All 25 individual points pooled	182.19 ± 4.02	0.0763 ± 0.0104	0.9890
Trial 4 omitted (four-trial mean)	184.17	0.0784	0.9921
Highest standard omitted	212.00	0.0469	0.9926
Trial 4 with actual concentrations	174.83	0.0680	0.9886
Log–log power-law fit of means	$\mathrm{OD} = 80.4\,C^{0.828}$		0.9979

9 Calculations

9.1 Stock and Dilution Concentrations

$$C_0 = \frac{m_{\text{protein}}}{V_{\text{water}}} = \frac{0.300\,\text{g}}{60\,\text{mL}} = \boxed{0.005\,00\,\text{g mL}^{-1}} \quad \text{(Trials 1–3)}$$

$$C_0 = \frac{0.299\,\text{g}}{60\,\text{mL}} = \boxed{0.004\,983\,\text{g mL}^{-1}} \quad \text{(Trial 4)}$$

$$C_0 = \frac{0.150\,\text{g}}{30\,\text{mL}} = \boxed{0.005\,00\,\text{g mL}^{-1}} \quad \text{(Trial 5)}$$

Each transfer combines 2.00 mL of sample with 2.00 mL of water:

$$C_{1:2} = 0.005\,00\,\mathrm{g\,mL^{-1}}\left(\frac{2.00}{2.00+2.00}\right) = \boxed{0.002\,50\,\mathrm{g\,mL^{-1}}},$$
$$C_{1:4} = \boxed{0.001\,25\,\mathrm{g\,mL^{-1}}}, \quad C_{1:8} = \boxed{0.000\,625\,\mathrm{g\,mL^{-1}}}, \quad C_{1:16} = \boxed{0.000\,312\,5\,\mathrm{g\,mL^{-1}}}.$$

9.2 Mean, Standard Deviation, and RSD

For the highest standard (0.984, 0.980, 0.980, 0.912, 0.971):

$$\bar{x} = \frac{0.984+0.980+0.980+0.912+0.971}{5} = \boxed{0.9654},$$
$$s = \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}} = \boxed{0.0302}, \qquad \mathrm{RSD} = \frac{0.0302}{0.9654}\times 100 = \boxed{3.1\%}.$$

9.3 Regression of the Mean Data

With $\bar{C} = 0.001\,937\,5\,\mathrm{g\,mL^{-1}}$ and $\overline{\mathrm{OD}} = 0.42932$:

$$S_{xx} = \sum(C_i-\bar{C})^2 = 1.4531\times 10^{-5}, \qquad S_{xy} = \sum(C_i-\bar{C})(\mathrm{OD}_i-\overline{\mathrm{OD}}) = 2.6474\times 10^{-3},$$
$$m = \frac{S_{xy}}{S_{xx}} = \boxed{182.19\ \mathrm{mL\,g^{-1}}}, \qquad b = 0.42932-(182.19)(0.0019375) = \boxed{0.0763}.$$

The residual standard deviation is $s_{y/x} = 0.0367$, so

$$s_m = \frac{0.0367}{\sqrt{1.4531\times 10^{-5}}} = 9.62, \qquad s_b = 0.0367\sqrt{\tfrac{1}{5}+\tfrac{(0.0019375)^2}{1.4531\times 10^{-5}}} = 0.0248,$$
$$\mathrm{LOD} = \frac{3.3(0.0367)}{182.19} = 0.000\,66\,\mathrm{g\,mL^{-1}}, \qquad \mathrm{LOQ} = \frac{10(0.0367)}{182.19} = 0.0020\,\mathrm{g\,mL^{-1}}.$$

9.4 Back-Calculation and Recovery

For the lowest standard, with OD = 0.0962:

$$C_{\mathrm{calc}} = \frac{0.0962-0.0763}{182.19} = 0.000\,109\,\mathrm{g\,mL^{-1}}, \qquad \mathrm{Recovery} = \frac{0.000109}{0.0003125}\times 100 = \boxed{34.9\%}.$$

9.5 Outlier Check on the Highest Standard

The lowest value (0.912, Trial 4) lies 0.053 below the mean of 0.9654. Grubbs' statistic is $G = (0.9654-0.912)/0.0302 = 1.77$, which exceeds the critical value $G_{\mathrm{crit}} = 1.715$ ($n = 5$, $\alpha = 0.05$). Dixon's ratio is $Q = (0.971-0.912)/(0.984-0.912) = 0.82$, which exceeds $Q_{\mathrm{crit}} = 0.710$ at 95% confidence. The value is therefore statistically flagged as a possible outlier, but it was retained (Section 10).

9.6 Illustrative Use of the Curve

If an unknown had $\mathrm{OD}_{595} = 0.500$, the curve would give

$$C_{\text{unknown}} = \frac{0.500 - 0.0763}{182.19} = 0.002\,33\,\mathrm{g\,mL^{-1}},$$

multiplied by the dilution factor if the sample had been diluted. This example is illustrative only; no unknown was measured.

10 Discussion

Trend and hypothesis. OD_{595} increased steadily with concentration in every trial: the mean rose from 0.0962 at $0.000\,312\,5\,\mathrm{g\,mL^{-1}}$ to 0.9654 at $0.005\,\mathrm{g\,mL^{-1}}$, supporting the hypothesis. In terms of the bacterial-growth analogy used in the lab, the concentrated stock resembles a dense, stationary-phase culture, the 1:2 and 1:4 standards resemble mid- and early-log growth, and the most dilute standards resemble the lag phase.

Reproducibility. The five trials agreed well. The slopes had a relative standard deviation of 2.7%, and the RSD of individual OD readings was 3–6% for the four more concentrated standards. The precision worsened to 8.8% at the lowest concentration. This is expected because each dilution compounds the pipetting error of the previous one, and because the same absolute noise of about 0.01 OD is a larger fraction of a small signal. Trial 4 gave the lowest OD at every concentration and the lowest slope (174 versus 180–187), and its stock reading of 0.912 was flagged by both Grubbs' and Dixon's tests at 95% confidence. Possible causes include a bubble, a smudged cuvette face, incomplete mixing, or particle settling before the sample was drawn. Because no procedural error was documented and the manual says to record every replicate and not to discard unfavorable results, the point was kept; omitting Trial 4 changes the slope by only about 1% (Table 5).

Is the relationship really linear? The high R^2 values (0.989–0.994) can be misleading. The residuals of the mean fit are not random (Figure 4a): they are negative at both ends and positive in the middle, which is the signature of curvature. Several observations agree:

- Each halving of concentration lowers OD by a factor of only 1.70–1.91 rather than 2, so OD is not strictly proportional to C.
- A log–log fit gives $\mathrm{OD} \propto C^{0.83}$ with $R^2 = 0.9979$, a better description than the straight line, and the linear fit bends away from the data at low C (Figure 4b).
- The linear intercept, 0.0763 ± 0.0104 (pooled, $n = 25$), is significantly different from zero ($t \approx 7.4$), yet the blank was zeroed and a physically meaningful response at $C = 0$ must be zero. The positive intercept is therefore best understood as an artifact of fitting a curved response with a straight line, not as real signal. The power-law fit passes essentially through the origin.
- Back-calculating concentrations from the line recovers the upper standards within 6–14% but the lowest standard at only 35% (Table 4). The estimated LOQ of $0.0020\,\mathrm{g\,mL^{-1}}$ is also higher than the lowest standards, although it is inflated by this lack of fit and should be viewed as a conservative estimate.

The sub-linear response is plausible for a scattering suspension: at higher particle density, particles shadow one another and multiple scattering becomes important, so each added particle contributes slightly less additional attenuation. Removing the highest standard raises the slope to 212 and lowers the intercept to 0.047, which confirms that the fit parameters depend on which part of the range is used. The straight-line

curve should therefore be applied only to the middle of the range, roughly 0.0006–0.005 g mL^{-1}, and never extrapolated. An unknown with OD above about 1.0 should be diluted and remeasured, as recommended in the manual, and a nonlinear (power-law) calibration would be more appropriate if the lowest standards matter.

Effect of technique on data quality. *Blanking* with the same solvent and cuvette type removes solvent and cuvette effects; a wrong or dirty blank would shift every reading by the same offset. *Wavelength* must be held at 595 nm, because turbidity depends on wavelength and other settings would not be comparable. *Cuvette condition and orientation* matter because fingerprints, scratches, droplets, and bubbles scatter light. *Mixing* immediately before measurement matters because the protein particles settle, so a poorly mixed sample under- or over-represents the intended concentration. *Dilution technique* (clean tip for each transfer, adding water first, mixing each tube before the next transfer) determines how accurately the nominal concentrations are realized.

Systematic and random errors. A systematic error example is a smudged or scratched cuvette face (or a bubble in the light path) that is present in every measurement of a series: it scatters extra light and biases OD *upward*. A consistently biased pipette would likewise shift all concentrations in the same direction, and the error compounds down the dilution series. A random error example is particle settling or incomplete mixing from sample to sample, which makes OD readings fluctuate unpredictably; settling before sampling would tend to *lower* the OD of that sample, as may have occurred in Trial 4.

11 Conclusion

Five trials of a protein-powder serial dilution measured at 595 nm showed that OD increases systematically with particle concentration, supporting the hypothesis. The mean standard curve was $\mathrm{OD}_{595} = 182.19C + 0.0763$ ($R^2 = 0.9917$), and the five trials agreed in slope to 2.7% RSD. Several procedures worked well: serial 1:2 dilutions with pre-filled water and a clean tip for each transfer, careful blanking, and consistent cuvette handling gave reproducible results with all stock ODs near the 0.90–1.10 target. The least reliable steps were measurement of the most dilute standards (8.8% RSD) and the Trial 4 stock, which read low. A residual analysis showed that the response is slightly sub-linear ($\mathrm{OD} \propto C^{0.83}$), that the nonzero intercept is a fitting artifact, and that the linear equation should be used only between about 0.0006 and 0.005 g mL^{-1}. To improve the next trial, each standard should be mixed immediately before it is read, the cuvette should be rewiped and rechecked for bubbles at every reading, and duplicate readings should be taken for each standard; adding lower-concentration standards and fitting a power law would improve accuracy at low turbidity. No unknown OD was recorded, so no unknown concentration could be calculated.

Waste Disposal and Cleanup

Protein suspensions were disposed of as permitted by the instructor, and contaminated tips and tissues were placed in the designated waste. Cuvettes were washed and dried without scratching the clear optical faces. The balance, pipetting area, and spectrophotometer workspace were wiped down and all equipment was returned to its storage location.

References

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3. Miller, J. N.; Miller, J. C. *Statistics and Chemometrics for Analytical Chemistry*, 6th ed.; Pearson: Harlow, 2010.